

SECTION | 3

Relational Neurobiology in IFIO

NEUROBIOLOGY AND EARLY LIFE EXPERIENCES*

In utero and during infancy (i.e., approximately the first 18 months of life), the brain experiences, learns, and “remembers” only through perception, sensation, emotion, and a sense of “knowing.” Because infants have not yet developed the capacity for language, the brain has no storytellers at this developmental phase, which means our earliest memories are nonverbal and don’t follow a linear storyline. Memories that form at this stage are known as implicit memories (Badenoch, 2008; Siegel, 1999). Around 18 months of age, we begin to acquire language and develop a form of autobiographical memory called explicit memory. With this new cognitive ability, our parts start telling stories and “making sense” of implicit learning that ascends into awareness.

During the early, implicit-memory-only phase of life, a child’s attachment style develops in their interactions with parents and other caregivers. That is, early relational experiences form a template upon which children base all future relationships (Schore & Schore, 2007). When these early relational experiences are suboptimal and challenge the child’s basic needs and sense of safety, long-term changes to brain circuits involved in emotion regulation, self-soothing, and social relatedness can result. Additionally, deficits in early caretaking affect the child’s ability to regulate their ANS and self-soothe when they are afraid, which impairs their ability to attune and make connections with other people (Cozolino, 2006). This child is at risk of developing an insecure attachment style—they might be predominantly avoidant, or they might crave attention and fear it both at once—which continues into adulthood.

A person’s unconscious, implicit memories of childhood wounds are stored in the amygdala, which is a primitive part of the brain that is deeply affected by social and emotional experiences early in life. When a present experience evokes emotional baggage from the past in the form of an implicit memory, that old experience can wash through the body as if it is happening in the here and now

(Badenoch, 2008). The amygdala believes there is a threat in real time, and the body's ANS activates, turning off our capacity for rational thinking. As long as the present is subsumed by the past, stressful interactions can easily toss individuals back in time—but they won't be aware of what's happening.

As Bonnie Badenoch wrote in *Being a Brain-Wise Therapist*, “So many couples are locked in a pattern of shifting into dysregulation, either toward disengagement and deadness or hyperarousal and chaos. When they see how these patterns have been engrained through lack of parental responsiveness, they can begin to develop a soothing, mindful awareness of their own state and that of their partner, immediately increasing regulation. Being able to let go of blaming the other for these disruptions makes room for mutual compassion” (2008, p. 282).

To put this in parts language, when clients' young exiles blend, they feel overwhelmed with sensory information from the past, which sends the inner system a terribly discouraging message: The past will never end. When deeply shaming attachment injuries are superimposed onto present interactions and the past feels relentlessly duplicated in the present, protectors naturally believe that their extreme views and vigilance are justified. As a result, couples fight relentlessly over who said or did what to whom. The irony is that their recall will never be accurate because both implicit learning and stress hormones distort explicit memory.

Since therapists can't know with certainty what someone said or did during high-stress moments, there is no point in helping couples try to decipher what “really” happened. While a protector's need for validation makes sense, we validate its feelings and confusion, but not strategies that cause further harm. Protectors get clarity and relief when the exile's original experience is witnessed, and enough healing takes place to jettison shaming beliefs (e.g., “I'm unlovable”).

We guide partners to break the habit of sparring over what really happened by introducing them to better alternatives, especially by helping them understand their neurobiology and by practicing unblending. In the process, we validate the good intentions of extreme protectors without endorsing their distorted views and guide partners in practicing new behaviors that change neurobiology, nourish confidence, and heal relationships. In couple therapy, secure attachments knit (or reknit) when the partner (outside) and the Self (inside) are available to extend kindness, attention, genuine curiosity, mature empathy, and loving compassion to wounded parts.

THE POLYVAGAL THEORY AND CO-REGULATION

The early view of the ANS posited that the nervous system comprised two branches—the sympathetic and parasympathetic branches—which served to activate and calm the body, respectively. However,

Stephen Porges expanded this view with the Polyvagal Theory (2007), which maintains that the nervous system has three branches that constantly respond to relational conditions and work together in a hierarchical fashion.

On the one hand, we have the *sympathetic nervous system*; on the other, we have the *parasympathetic nervous system*, which comprises a dorsal vagal branch and a ventral vagal branch. The sympathetic nervous system responds to cues of danger by triggering the fight-or-flight response, which mobilizes the body for action. In contrast, the dorsal vagus and ventral vagus are both parts of the parasympathetic nervous system. The *dorsal vagus* is an evolutionarily older branch of the parasympathetic nervous system. It responds to cues of extreme danger with freeze or collapse behaviors (Dana, 2018). In IFIO, we would say that protective parts who operate in synchrony with the dorsal vagal system numb or freeze us, taking us out of connection and awareness, signaling their sense that extreme protection is in order.

Conversely, the *ventral vagus*, which is a more recent parasympathetic pathway, responds to cues of safety and promotes social connection. In IFIO, we theorize that a ventral vagal state facilitates Self-leadership.

Polyvagal Theory also posits two concepts of particular relevance for couples: neuroception and co-regulation. Neuroception refers to the unconscious process by which we are constantly scanning the external environment for cues of safety or danger. Rather than being logical or verbal, neuroception is a somatic, emotional, and associative process. In neuroception, the amygdala matches current experiences with prior experiences relevant to safety, which causes the ANS to react with relaxation or activation.

Subsequently, the left hemisphere makes up a story about what is happening, and this narrative is informed by the degree to which someone is led by their parts or their Self. When individuals are parts-led, the ANS is more likely to be in charge and the past has a greater influence over the storyline. In contrast, when individuals are Self-led, they can slow down, evaluate the accuracy of their reaction, check with their partner regarding their experience, and choose whether to respond and how to act wisely.

According to Polyvagal Theory, connecting with others provides us with the neuroception of safety, and it is through the formation of trusting relationships that we can co-regulate into a ventral vagal state. Co-regulation involves the reciprocal regulation of our ANS in connection with others. It

creates a reassuring platform that signals safety and promotes further attachment. When a couple can co-regulate their nervous systems, they experience the pleasure of a self-reinforcing, co-created connection (Dana, 2018).

Importantly, a couple's concept of reciprocity in their relationship—the healthy, mutual give-and-take of support, love, and understanding—gives us more information about their dynamics and helps assess their ability to co-regulate in moments of stress (Dana, 2018). Of course, the reciprocity of giving in a couple will differ over time. In IFIO, we help partners keep noticing and assessing their ability to reciprocate.

NEUROBIOLOGY IN IFIO

We help couples by:

- Educating them on the ANS and normalizing their experience
- Teaching them to unblend from parts and helping each partner witness their relational trauma, which provides a contradictory, corrective experience for their exiles
- Giving the couple a disconfirming, reparative experience as they witness each other from the Self and hold compassion for their partner's wounds
- Providing education about the unconscious process of neuroception
- Using top-down/bottom-up interventions (Anderson, Sweezy, & Schwartz, 2017)
- Educating partners about co-regulation and reciprocity and conducting in-office exercises so partners can experience both
- Noticing co-regulation when it occurs spontaneously in the office between partners and enhancing the moment as much as possible

REGULATING THE AUTONOMIC NERVOUS SYSTEM

The following vignette illustrates self-regulation and co-regulation experiments to help a European-American couple, Dan and Cate, regulate their ANS in session. Dan is a transgender man who transitioned in his late twenties, and Cate is a cisgendered female who identifies as queer. They have been married for five years and together for seven. They sought couple therapy because both

were reacting intensely to real or perceived threats in conversation, after which repair and reconnection were proving difficult.

Therapist: How was your week? And what will be helpful for today's session?

Dan: We just do not get along. I can never get anything right. I'm always in the doghouse with Cate.

Cate: Yes! Everything is about you, by the way. What about me? I have needs too, Dan!

Dan: See? We're off and running! [*he rolls his eyes*]

Therapist: Okay. You say the two of you are struggling to get along, Dan. You have a part who feels it can't do anything right. And you, Cate, spoke of parts with unmet needs. These issues are causing friction in your relationship. And both of you feel frustrated. Have I got all that right?

Dan: Yes.

Cate: That's right!

Therapist: This is just where we left off last time, so let's continue. [*Dan nods*]

Cate: [*to Dan*] You're always angry and yelling, by the way. I can't deal with that.

Dan: You're so damn condescending—who wouldn't be angry?

Therapist: Let's take a minute to notice what's happening right now. Start with your bodies. Take a few deep breaths. [*after a moment*] I want to remind you about your nervous systems. Are you fighting, fleeing, or freezing right now?

Cate: Okay. I have a part that's fleeing big time. I can hear its voice saying, "Get me out of here!"

Dan: And I'm in my usual place—itching for a fight.

Therapist: And what's happening in your body right now?

Dan: I'm cranking up. I'm tight. I can't help it.

Therapist: Got it, Dan. You're angry and getting more agitated. And you don't feel like you can help yourself. Does that sound right?

Mirroring

Dan: Right.

Therapist: [*turning to Cate*] And your body?

Cate: I'm tense, but it's not like I am going to get up and run. Not now anyway. Breathing deeply helps.

Therapist: Great. If you can keep noticing your body and your parts, I want to spend a little time with Dan.

Cate: Fine with me.

Therapist: [*to Dan*] All this energy in your body makes sense to me. It sounds like you're feeling unsafe? Can you look at me?

*Offering co-regulation through eye contact
and tone of voice*

Therapist: [*continuing*] I have heard from you, Dan, that when you feel offended or threatened, you can get very angry, and the anger takes over your body. And sometimes it's hard for this angry part to unblend, as we say in IFS. It's hard to get yourself to the calm zone. And given your history, this makes complete sense to me. Is there more you want to say right now about how you're feeling?

*Validation and mirroring followed by an
invitation to say more about feelings*

Dan: I'm a little better. But I feel angry. Really angry.

Therapist: You are really angry.

Focusing on his feeling state

Dan: Right.

Therapist: And you see the anger as a part?

Dan: I do.

Therapist: Where do you feel this part in or around your body?

Bringing awareness back to the body

Dan: My chest.

Therapist: How do you feel toward it?

Checking for unblending

Dan: I feel like I *am* him. He's all there is right now.

Therapist: Can I speak to him?

Getting permission for direct access

Dan: Fine.

Therapist: [*turning to Cate*] Are you with us?

Staying connected with the partner

Cate: I am. I feel more relaxed knowing you are with Dan and his anger.

Therapist: [*using direct access*] So I'll talk to Dan's angry part. Can you hear me? [*Dan nods*]
Can you help me understand what's going on?

Dan: [*the angry part answers directly*] You don't understand what a weak, sniveling, incapable, spineless person Dan is! I am his only hope of staying alive.

Therapist: I see. You are strong and powerful?

Dan: [*the angry part*] He's alive, isn't he? He would probably be dead if it weren't for me. No one—and I mean no one—is going to threaten him ever again.

Therapist: And that includes those parts of Cate?

Dan: [*looks at Cate*] I guess it does.

Therapist: [*to Dan's Self*] Are you listening, Dan?

Dan: I am.

Therapist: How are you feeling toward this part now?

Checking for unblending

Dan: I recognize these words. I started saying this in my teens.

Therapist: You have a part who started protecting you in this way back then?

Dan: Yes. And it's all true. I might be dead without this part.

Therapist: What's happening in your body right now?

Checking his level of arousal

Dan: I feel calmer. But I still feel a lot of energy coursing around.

Therapist: I hear you. Lots of energy in your body. Can you handle it?

*Checking for unblending and getting
permission to continue*

Dan: I can.

Therapist: Are you available for more inquiry with this protective part?

Dan: I can try. The more I notice him, the more he changes. He was a lion, now he looks like a teenager.

Therapist: Does that make sense, or do you need to ask him about it?

Dan: No. It makes sense.

Therapist: When it feels right, please ask him whom he protects.

Dan: [long pause] Oh GOD! I don't want to go there.

Therapist: Because?

Dan: It's girl part. A young girl. Confused, terrorized.

Therapist: Who is afraid right now?

Dan: [raises his voice] Everyone is afraid of this girl! Do you not get this?

Therapist: Dan, I very much want to get this. What are they afraid of?

Dan: I will not be dragged back into the past. I left all that behind.

Therapist: So you have parts who feel threatened by her?

Dan: Yes.

Therapist: Visiting the past feels dangerous? I'm hearing that you have a part who is still trying to protect her, and you also have parts who want to exile her. Is that right?

Naming the polarized protectors regarding this exile

Dan: That's exactly right.

Therapist: Can you feel this dilemma in your body?

Dan: So incredibly much anxiety!

Therapist: We're going to help all these parts, Dan. But first I'm going to check in with Cate. [*Dan nods, and the therapist turns to Cate*] Are you staying present? Does any of this make sense to you?

Cate: Yeah. It's funny. I sensed this somehow. Something is being pushed down all the time. And Dan acts like I am a threat. I can be an asshole, but I'm not going to beat him up or anything.

Dan: [*looks up sharply and snaps*] Well it feels like you are! You criticize me constantly.

Therapist: What just happened, Dan?

Dan: The anger came back.

Therapist: Do you notice how quickly this part deploys when it feels you're being threatened? [*Dan nods*] And if we could do something to help this giant protector feel safer, would he need to react to Cate this quickly?

Inviting the part to receive support and be less reactive so the client has more choice in responding

Dan: It's true this part feels like he has no choice when he responds. So how would we change th

Therapist: We continue to help you understand three important things. First, this is a protector who activates without your conscious awareness—that is, without your permission. He takes over your body. Second, your reaction makes complete sense given your childhood. And, third, this part is operating on the assumption that Cate is more

dangerous than she really is. Since all this is true, we have several ways for you to help him unblend and regulate your nervous system, including healing the part this teenage boy protects.

Explaining the autonomic responses to perceived danger and the body's way of coping, and then offering an alternative

Dan: You mean the girl part?

Therapist: Exactly. What's happening inside you as I'm sharing all this?

Dan: I'm relaxing. I'm curious. When I'm calm, I can recognize that my reactions are often way bigger than the scenario would justify. But when I'm in it, I feel so justified! [*looks at Cate*] What do you think of this?

Cate: I am thinking about you, but I am also thinking about me and how quickly I react. It's a relief to know we have reasons for doing what we do—and that we might have a way out.

Therapist: You can also help each other in these moments. When we get triggered in a relationship, we often have the experience of feeling alone and unsafe. A couple of important strategies for changing this are learning to unblend and learning a safe and courageous way to communicate your feelings. Would you be interested?

Offering the tool of courageous communication (see [Section 2](#))

Cate: Well this is one of our big goals, isn't it, Dan?

Dan: Yes, it is.

After the therapist helped Cate and Dan address their protectors' fears so these parts could unblend, the session went on.

Therapist: Dan, I wonder if you would like to speak on behalf of any of the parts we've spent time with today?

Dan: I think Cate knows them.

Therapist: She might know them directly, but does she know them through you?

Cate: [nods] We're on the receiving end of each other's paranoid, mean parts.

Validating and taking responsibility, which creates reciprocity and facilitates ANS regulation (Dana, 2018)

Dan: Okay. I'll give it a go. For today, Cate, this is what I know. I can be angry and mean. That's part. He gets triggered when you are critical.

Therapist: Can I restate that? [Dan nods] Okay. He gets triggered when you feel criticized.

Dan: Well, okay. Sometimes I experience you that way, Cate.

Cate: I can own that. I can be critical.

Dan: Okay. So it's like this volcano erupts in my body. It's this part who gets enraged and indignant. What I'm hearing is that he is protecting another part who got beat up when I was young. That message was "You are unacceptable." This message is "Never again!" Let's talk about expressing emotion, for example. My family could never understand what was happening with me. No teachers, no friends. No one understood. The bullying was relentless. I was tortured a few times. Did you know I was raped when I was 14? [Dan begins to cry]

Therapist: Is it okay to feel this right now?

Asking permission to continue

Dan: [nods and continues to cry] It's the part who got hurt. She was so scared and alone! She was brutalized.

Therapist: Can we be with her now?

Dan: I am with her. But I feel these other parts want to bury her.

Therapist: Does that make sense to you?

Checking for unblending

Dan: It does.

Therapist: Is she aware of you?

Dan: She is.

Therapist: What makes sense right now given her needs?

Dan: I want Cate to know about her.

Therapist: Cate, are you available to hear about this part of Dan?

Cate: I am! I've always wondered about this, Dan. It seems like you've had to cut a piece off. [*speaking in a soft tone of voice*] Can I move closer to you?

At this point, Cate moved toward Dan and reached out to touch his knee. Spontaneous touch is a bid for connection and co-regulation.

Dan: I like that. I feel you. She was so trapped, so frantic. Trying to fit in made me... made this part of me sick. My mother could not deal with me... her.

Therapist: How are you experiencing her?

Dan: She looks like a little boy actually. Or should I say, how we identify little boys?

Therapist: How do you feel toward her now?

Dan: Really, really sad.

Therapist: [*to Cate*] What moves you?

Cate: Big sadness. I've heard some stories of course, but not this.

Therapist: We don't have much time left today. I am wondering, Dan, if you could stay inside with this part a little longer?

Dan: [*closes his eyes*] I can tell you this isn't easy given the resistance I feel toward her.

Therapist: Other parts?

Dan: Yes. They are afraid of her.

Therapist: Is there anything you want to say to them?

Inviting the client's Self to respond to an inner dilemma

Dan: Yes. I need to do this. The past haunts us. She needs me. It's time.

Therapist: And how do they respond?

Dan: They're okay right at this moment. But I know this is just the beginning. Right now, there's not a lot of confidence about me doing this.

Therapist: Given that this is the beginning and the process is new, does their doubt make sense to you?

Dan: I get it. I'll need to show up.

Therapist: Let's go back and help the girl find a place that will feel safe and comfortable until we return.

Dan: She wants to stay here, with us.

Therapist: How is that for you?

Dan: Good.

Therapist: Cate, is there anything you want to say to Dan or to this part? From your heart.

Asking the partner to provide validation, which helps diminish arousal (Fruzzetti & Worrall, 2010).

Cate: [after a pause] I appreciate you being willing to go there. You do make sense. I wish these things hadn't happened to you. And I want you to know that I know our problems aren't all you, even though I sometimes say you are the problem. The way I see it now, our relationship can only get better if we unwind our miserable childhoods.



This session was complex for a few reasons. Initially, both partners were autonomically aroused. While Cate was able to self-regulate, Dan needed help. Therefore, the therapist focused on supporting him while being careful not to abandon Cate or lose sight of holding them as a couple. Over the course of the session, the therapist educated them about their autonomic nervous systems, helped Dan reach an optimal state of arousal (using the top-down interventions of validation, mirroring, and inviting him to speak for feelings), and then helped him make a U-turn so he could attend internally.

Inside, Dan learned that an angry teenage boy was protecting a terrified girl who looked like a boy and had been exiled internally as well as externally. His system continued to view her as a threat. By regulating his ANS, Dan was able to exhibit a Self-led challenge to this shunning, and he

included Cate in his process. He saw that the girl needed rescue and wisely understood that the fears of his other parts would get in the way. This session illustrates how external validation reassures and internal reconnection soothes, helping individuals and couples move from an initial state of dysregulated disconnection toward optimal calm and presence.

When couples like Dan and Cate come to therapy feeling dysregulated—with their protective systems in chronic overdrive—our job is to help them recapture trust. When they feel despair, our job is to be patient and persistent. The steps of IFIO are all designed to support couples in developing safety and trust. The goal is to help them be aware of but not blended with their parts. The more unblended their parts, the more regulated their nervous system. The more regulated the nervous systems of both partners, the more likely they will support instead of abandon each other in times of conflict and distress. A well-regulated nervous system invites relational synchrony, inside and out.

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